BLOCKCHAIN Experiment No: 9

Harshvardhan Patil
D20A/22

Aim

Create an ERC20 Token in Remix IDE: write the EduCoin (EDC) token contract in Solidity, compile it, connect Remix IDE to MetaMask on the Sepolia testnet, deploy the token, and verify the deployment by testing its ERC20 functions.

Tasks Performed

- Completed all six chapters of the LearnETH Solidity ERC20 course.
- Set up the MetaMask wallet in the browser and enabled the Sepolia testnet.
- Funded the wallet with Sepolia ETH from the Google Cloud Web3 faucet.
- Created the EduCoin.sol smart contract in Remix IDE.
- Compiled the contract with Solidity compiler 0.8.18 (OpenZeppelin Contracts 4.4.0).
- Connected Remix IDE to MetaMask (Browser Extension environment) on Sepolia.
- Deployed the EduCoin contract to the Sepolia testnet.
- Tested the ERC20 functions: name, symbol, decimals, totalSupply and balanceOf.
- Imported the EDC token into MetaMask and verified the contract on Sepolia Etherscan.

Tools and Libraries Used

- Remix IDE (remix.ethereum.org)
- MetaMask browser wallet
- Solidity 0.8.18
- OpenZeppelin Contracts 4.4.0 (ERC20 implementation)
- Sepolia testnet (Chain ID 11155111)
- Sepolia Etherscan (sepolia.etherscan.io)
- Google Cloud Web3 faucet (Sepolia ETH)

Theory

ERC20 is the standard interface for fungible tokens on Ethereum. It defines a common set of functions (totalSupply, balanceOf, transfer, approve, allowance, transferFrom) and events (Transfer, Approval) so that wallets, exchanges and dApps can interact with any compliant token. OpenZeppelin's ERC20 contract provides a secure, audited implementation of this standard: the developer only declares the token name and symbol and decides the initial supply, while inheritance supplies the full standard behaviour including 18 decimals by default.

Deploying to the Sepolia testnet (instead of Ethereum mainnet) lets the contract run on a public blockchain with real consensus and a block explorer, while gas is paid in SepoliaETH obtained free

from a faucet - so no real money is ever spent. Remix IDE's 'Browser Extension' environment routes deployment transactions through MetaMask, which signs them with the user's wallet.

Program: EduCoin.sol

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.3;

import "@openzeppelin/contracts@4.4.0/token/ERC20/ERC20.sol";

contract EduCoin is ERC20 {

    constructor() ERC20("EduCoin", "EDC") {
        _mint(msg.sender, 1000 * 10 ** decimals());
    }
}
```

Code Explanation

- pragma solidity ^0.8.3; - Declares the Solidity compiler version compatibility.
- import .../ERC20.sol; - Imports the OpenZeppelin ERC20 implementation (v4.4.0).
- contract EduCoin is ERC20 - Creates EduCoin inheriting full ERC20 functionality.
- ERC20("EduCoin", "EDC") - Sets the token name (EduCoin) and symbol (EDC).
- _mint(msg.sender, 1000 * 10 ** decimals()); - Mints the initial supply of 1000 EDC to the deploying wallet.
- decimals() - Returns the default precision of 18 decimals, so 1000 EDC = 1000 * 10^18 base units.

Output

Figure 1: MetaMask wallet unlocked on Sepolia.

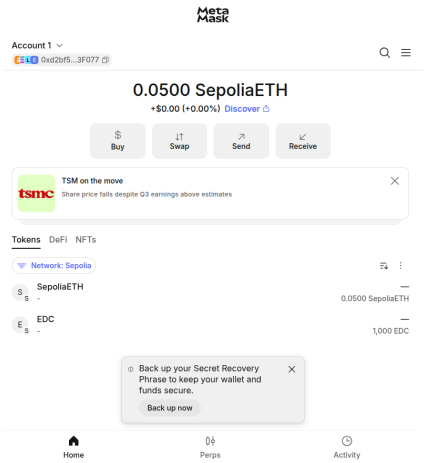
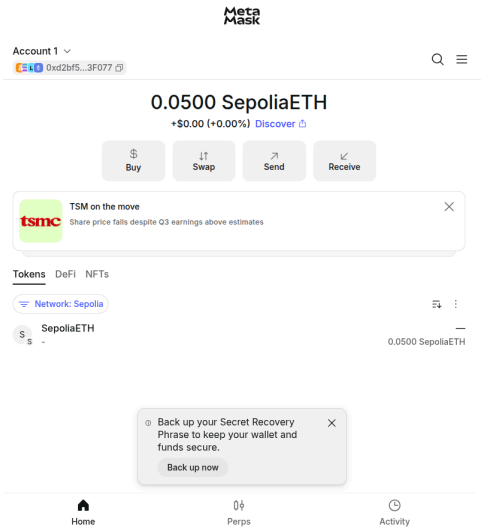


Figure 2: Sepolia testnet selected in MetaMask.



The Sepolia testnet (Chain ID 11155111) was selected from the network picker's Testnets section. The wallet address is 0xd2bf51666c0ec239c5bfcf6c2fc520a02b63f077 and the balance reads 0.0500 SepoliaETH.

Figure 3: Sepolia ETH received (faucet funding).

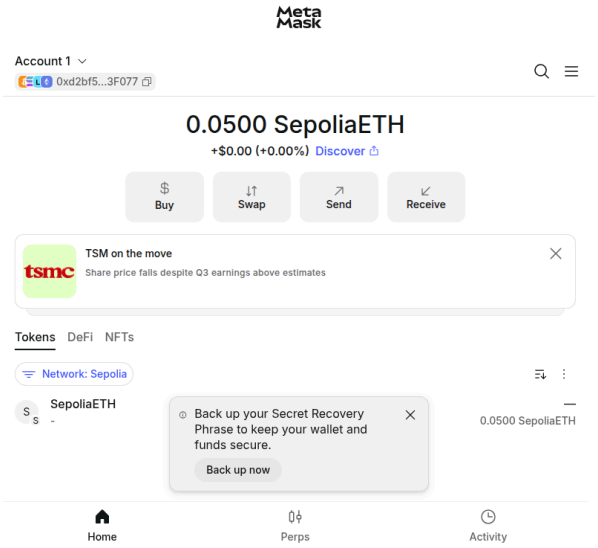


Figure 4: EduCoin.sol source code in Remix IDE.

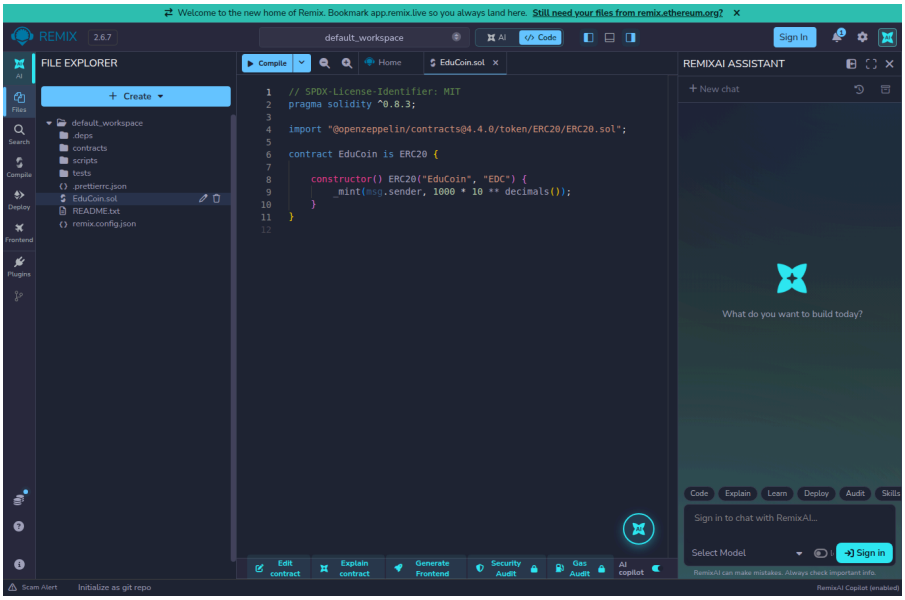


Figure 5: Successful compilation with Solidity 0.8.18.

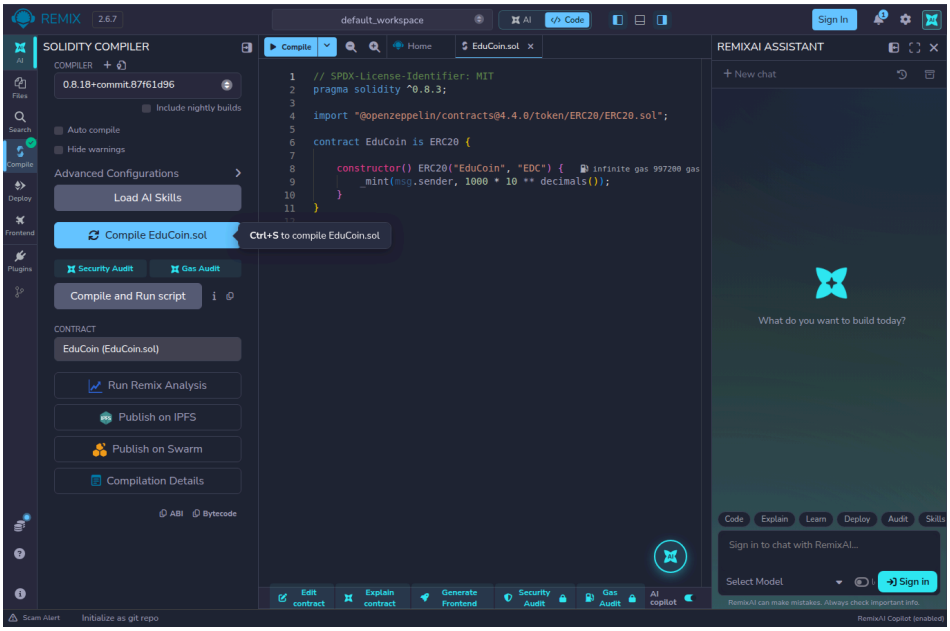


Figure 6: Remix connected to MetaMask on Sepolia.

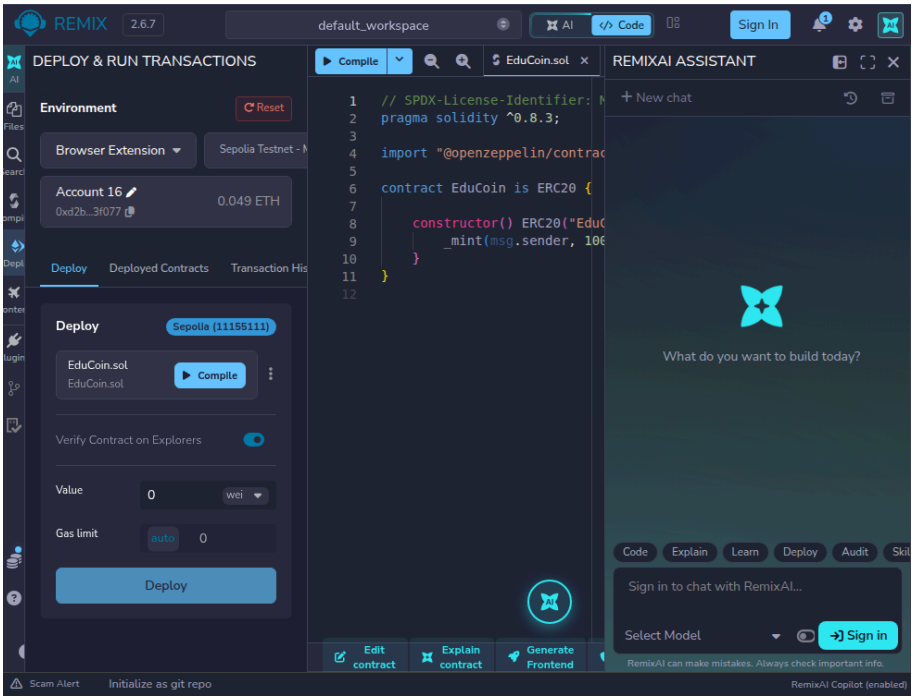


Figure 7: MetaMask connection approval for Remix IDE.

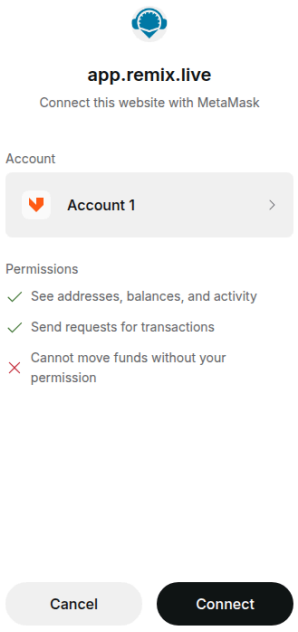


Figure 8: Deployed contract loaded in Remix (Add Contract).

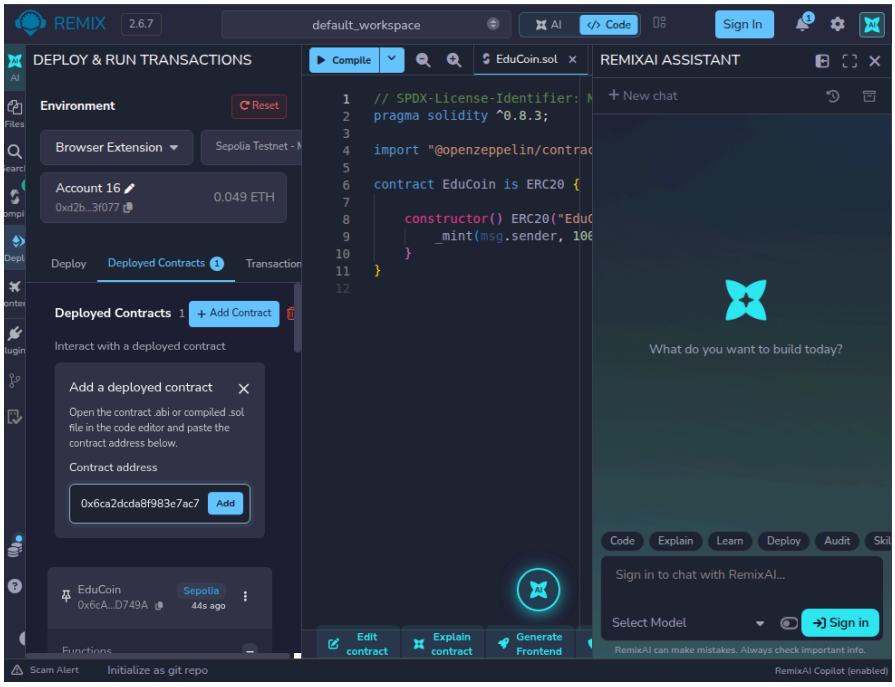


Figure 9: name() and symbol() return EduCoin / EDC.

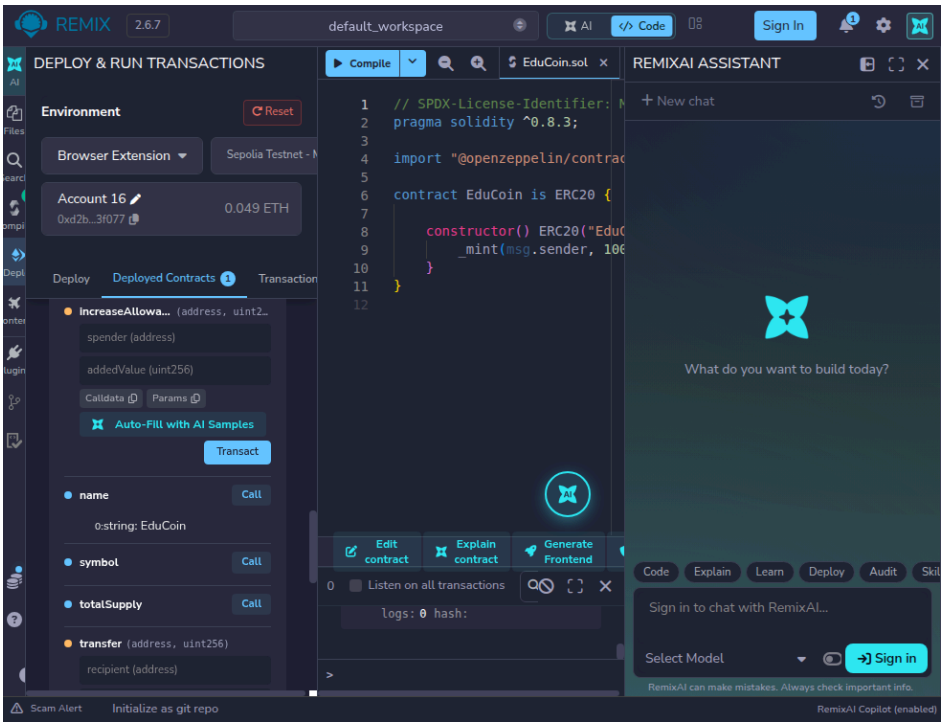
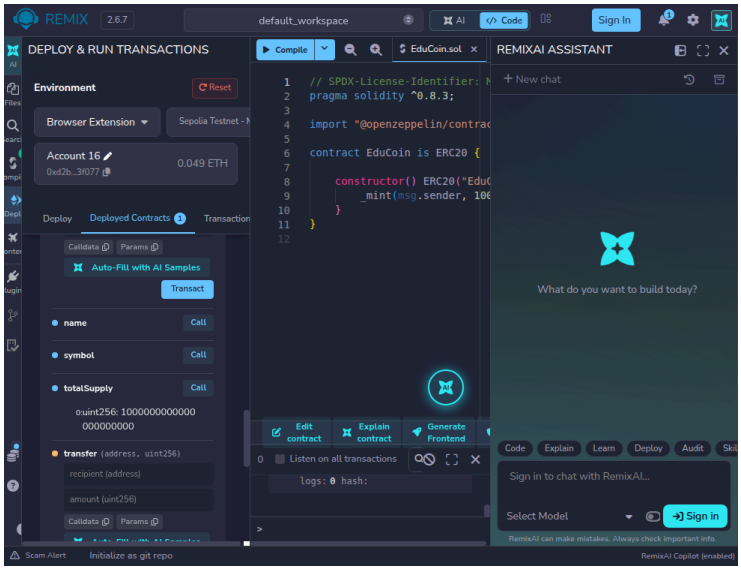
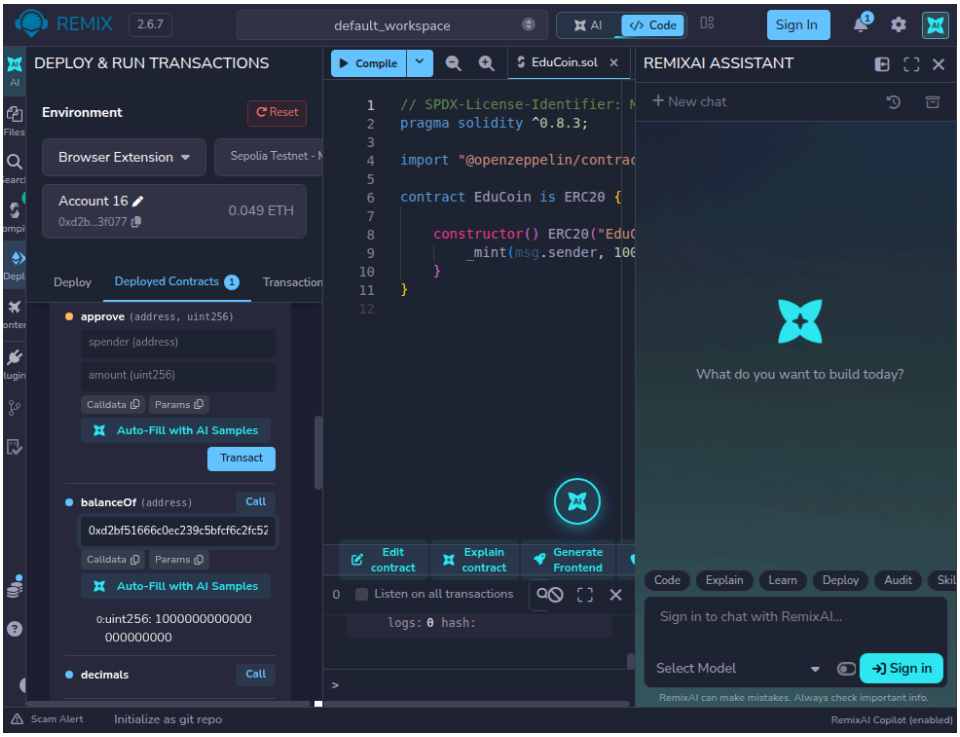


Figure 10: decimals() and totalSupply() verified.



Calling decimals() returned '0: uint8: 18' (the OpenZeppelin ERC20 default). Calling totalSupply() returned '0: uint256: 1000000000000000000', which equals 1000 * 10^18 - the 1000 EDC minted in the constructor. Both match the expected ERC20 behavior.

Figure 11: balanceOf(deployer) shows 1000 EDC.



Calling balanceOf(0xd2bf51666c0ec239c5bfcf6c2fc520a02b63f077) - the deployer wallet - returned '0: uint256: 1000000000000000000', confirming the full 1000 EDC initial supply was minted to the deployer as coded in the constructor.

Figure 12: EduCoin (EDC) visible in MetaMask.

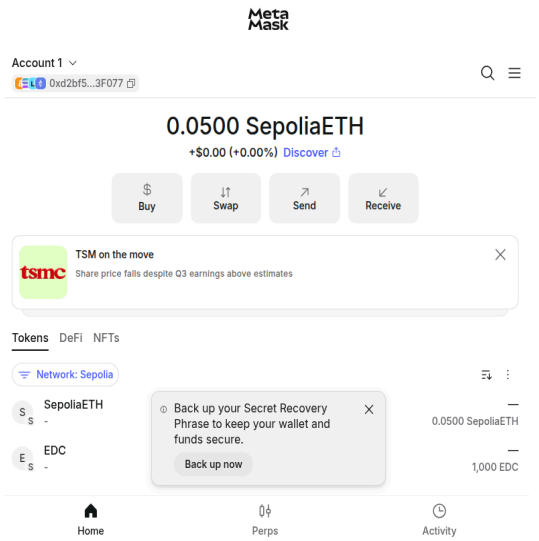
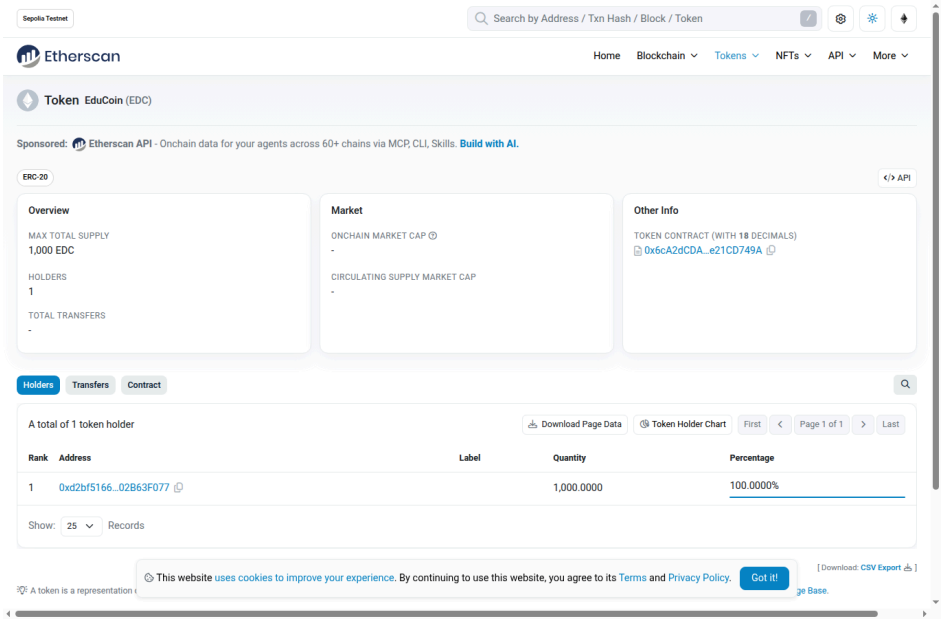


Figure 13: Contract live on Sepolia Etherscan.



Conclusion

Successfully created and deployed an ERC20 token named EduCoin (EDC) using the Solidity programming language and Remix IDE. The OpenZeppelin ERC20 library was used to implement the standard token functionality. MetaMask was integrated with Remix IDE and the Sepolia testnet was used for smart contract deployment. An initial supply of 1000 EDC tokens was minted and allocated to the deployer's wallet. The token functions, including name, symbol, decimals, totalSupply.